

# HENDERSON EXECUTIVE, NV, USA

WMO: 722096

Lat: 35.976N

Lon: 115.133W

Elev: 749

StdP: 92.64

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 53127

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -2.1       | -0.3 | -20.2                         | 0.7 | 13.5 | -17.9 | 0.8 | 12.5 | 14.1                  | 14.3 | 12.3 | 12.9 | 2.4                   | 160  | 0.541 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.9                   | 42.5            | 18.6 | 41.3 | 18.3 | 40.1 | 18.0 | 22.0                | 34.1 | 21.4 | 34.0 | 20.7 | 33.9 | 4.6                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.7                            | 14.9 | 26.3 | 17.5 | 13.7 | 27.7 | 16.1 | 12.5 | 29.0 | 68.0          | 33.7 | 65.6 | 33.9 | 63.0 | 33.7 | 24.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 12.2          | 10.9 | 9.2 | -4.7                       | 44.8 | 1.8                | 1.2 | -6.0                                               | 45.7 | -7.0       | 46.4 | -8.1       | 47.1 | -9.4       | 48.0 |     |     |
| (5)               |      |     | -7.4                       | 23.0 | 1.5                | 0.7 | -8.5                                               | 23.5 | -9.3       | 24.0 | -10.1      | 24.4 | -11.2      | 24.9 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.2   | 8.4   | 10.8  | 15.2  | 19.0  | 23.6  | 30.7  | 33.3  | 32.0  | 27.7  | 20.1  | 13.5  | 7.6   | (6)  | (7)  |
|                                                                        | DBStd       | 9.60   | 3.18  | 3.82  | 3.94  | 4.10  | 4.43  | 3.59  | 2.60  | 2.48  | 3.62  | 3.83  | 4.04  | 3.21  | (7)  | (8)  |
|                                                                        | HDD10.0     | 208    | 67    | 34    | 5     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 14    | 87    | (8)  | (9)  |
|                                                                        | HDD18.3     | 1189   | 309   | 210   | 113   | 40    | 9     | 0     | 0     | 0     | 1     | 24    | 149   | 334   | (9)  | (10) |
|                                                                        | CDD10.0     | 3933   | 16    | 58    | 165   | 272   | 423   | 620   | 721   | 683   | 531   | 313   | 118   | 12    | (10) | (11) |
|                                                                        | CDD18.3     | 1872   | 0     | 1     | 15    | 62    | 173   | 370   | 463   | 425   | 282   | 78    | 4     | 0     | (11) | (12) |
|                                                                        | CDH23.3     | 27784  | 0     | 7     | 188   | 729   | 2258  | 5814  | 7578  | 6622  | 3727  | 801   | 60    | 0     | (12) | (13) |
|                                                                        | CDH26.7     | 17275  | 0     | 0     | 43    | 266   | 1162  | 3821  | 5214  | 4349  | 2143  | 273   | 4     | 0     | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 4.1    | 3.2   | 3.7   | 4.5   | 5.0   | 4.9   | 5.0   | 4.4   | 4.3   | 3.9   | 3.7   | 3.3   | 3.1   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 252    | 25    | 35    | 21    | 11    | 7     | 6     | 34    | 27    | 25    | 19    | 12    | 28    | (15) | (16) |
|                                                                        | PrecMax     | 353    | 93    | 103   | 103   | 37    | 30    | 20    | 82    | 70    | 80    | 54    | 46    | 107   | (16) | (17) |
|                                                                        | PrecMin     | 162    | 1     | 2     | 1     | 0     | 0     | 0     | 5     | 3     | 1     | 0     | 0     | 1     | (17) | (18) |
|                                                                        | PrecStd     | 63     | 29    | 29    | 22    | 10    | 7     | 5     | 22    | 17    | 18    | 15    | 11    | 25    | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 20.1  | 24.3  | 30.6  | 34.5  | 38.5  | 43.8  | 44.5  | 42.4  | 40.1  | 34.0  | 27.3  | 19.7  | (19) | (20) |
|                                                                        |             | MCWB   | 8.7   | 10.7  | 12.9  | 14.4  | 15.6  | 18.5  | 19.3  | 18.9  | 18.1  | 15.6  | 12.3  | 9.7   | (20) | (21) |
|                                                                        | 2%          | DB     | 18.2  | 22.3  | 27.6  | 31.9  | 36.4  | 41.9  | 43.1  | 41.2  | 38.4  | 31.6  | 25.1  | 17.6  | (21) | (22) |
|                                                                        |             | MCWB   | 7.7   | 9.8   | 11.9  | 13.2  | 14.9  | 17.8  | 19.0  | 18.7  | 17.6  | 14.7  | 11.7  | 8.3   | (22) | (23) |
|                                                                        | 5%          | DB     | 16.6  | 20.5  | 25.3  | 29.7  | 34.9  | 40.7  | 41.8  | 40.2  | 37.2  | 29.7  | 23.3  | 15.8  | (23) | (24) |
|                                                                        |             | MCWB   | 7.1   | 9.1   | 10.8  | 12.6  | 14.4  | 16.9  | 18.7  | 18.5  | 17.4  | 14.1  | 10.7  | 7.4   | (24) | (25) |
|                                                                        | 10%         | DB     | 15.0  | 18.6  | 23.3  | 27.6  | 32.9  | 39.3  | 40.4  | 39.1  | 35.8  | 27.9  | 21.3  | 14.0  | (25) | (26) |
|                                                                        |             | MCWB   | 6.4   | 8.4   | 10.1  | 11.8  | 13.8  | 16.3  | 18.6  | 18.3  | 17.0  | 13.2  | 10.1  | 6.4   | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 11.3  | 12.9  | 13.6  | 15.3  | 16.8  | 20.6  | 23.0  | 22.9  | 21.6  | 18.4  | 13.9  | 11.3  | (27) | (28) |
|                                                                        |             | MCDB   | 15.2  | 19.4  | 27.1  | 32.8  | 34.9  | 38.6  | 35.3  | 34.8  | 31.2  | 27.1  | 24.6  | 17.3  | (28) | (29) |
|                                                                        | 2%          | WB     | 9.6   | 10.9  | 12.6  | 14.1  | 15.7  | 19.3  | 22.2  | 22.0  | 20.4  | 16.9  | 12.3  | 10.0  | (29) | (30) |
|                                                                        |             | MCDB   | 15.3  | 18.8  | 25.4  | 29.4  | 33.8  | 38.7  | 34.8  | 33.5  | 33.4  | 26.7  | 23.3  | 15.2  | (30) | (31) |
|                                                                        | 5%          | WB     | 8.3   | 10.0  | 11.6  | 13.1  | 15.0  | 18.3  | 21.6  | 21.4  | 19.5  | 15.6  | 11.4  | 8.7   | (31) | (32) |
|                                                                        |             | MCDB   | 14.7  | 18.6  | 23.7  | 27.4  | 32.5  | 37.4  | 34.6  | 33.4  | 32.3  | 25.7  | 21.8  | 14.1  | (32) | (33) |
|                                                                        | 10%         | WB     | 7.2   | 9.0   | 10.6  | 12.3  | 14.3  | 17.2  | 21.0  | 20.7  | 18.7  | 14.6  | 10.4  | 7.2   | (33) | (34) |
|                                                                        |             | MCDB   | 13.5  | 17.9  | 22.0  | 26.1  | 31.2  | 36.3  | 34.3  | 33.3  | 31.0  | 25.1  | 20.2  | 12.9  | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.7  | 12.1  | 13.0  | 13.4  | 13.5  | 14.3  | 12.9  | 13.0  | 13.7  | 13.1  | 12.5  | 10.8  | (35) | (36) |
|                                                                        |             | MCDBR  | 13.9  | 14.6  | 15.5  | 15.7  | 15.3  | 15.1  | 14.1  | 14.0  | 14.8  | 14.8  | 14.6  | 12.9  | (36) | (37) |
|                                                                        | 5% WB       | MCWBR  | 7.4   | 7.2   | 6.8   | 6.5   | 5.7   | 5.4   | 5.0   | 5.2   | 5.6   | 6.4   | 6.9   | 7.0   | (37) | (38) |
|                                                                        |             | MCWBR  | 11.3  | 12.7  | 14.3  | 14.4  | 14.4  | 13.9  | 11.7  | 11.9  | 12.5  | 11.9  | 13.4  | 10.5  | (38) | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.262 | 0.266 | 0.280 | 0.301 | 0.318 | 0.315 | 0.353 | 0.344 | 0.311 | 0.288 | 0.265 | 0.261 | (40) | (41) |
|                                                                        | taud        |        | 2.625 | 2.628 | 2.564 | 2.482 | 2.456 | 2.476 | 2.453 | 2.468 | 2.545 | 2.597 | 2.632 | 2.633 | (41) | (42) |
|                                                                        | Ebn at Noon |        | 961   | 991   | 997   | 983   | 963   | 961   | 922   | 927   | 949   | 951   | 948   | 939   | (42) | (43) |
|                                                                        | Edh at Noon |        | 71    | 80    | 94    | 107   | 112   | 110   | 112   | 108   | 94    | 82    | 70    | 67    | (43) | (44) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 3.09  | 4.05  | 5.56  | 6.96  | 7.89  | 8.54  | 7.52  | 7.00  | 6.18  | 4.66  | 3.48  | 2.74  | (44) | (45) |
|                                                                        | RadStd      |        | 0.24  | 0.35  | 0.26  | 0.29  | 0.38  | 0.34  | 0.39  | 0.35  | 0.25  | 0.27  | 0.14  | 0.20  | (45) | (46) |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
|                         |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (15 neighbors) |  | +0.44 | N/A     | -1.60  | N/A     | N/A   | N/A   | N/A         | N/A     | +159    | N/A     |

Nomenclature: See separate page